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Angle-constrained formation control under directed non-triangulated sensing graphs[☆]Kun Li, Zhixi Shen, Gangshan Jing^{*}, Yongduan Song

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ABSTRACT

Angle-constrained formation control has attracted much attention due to the advantage that inter-edge angles are invariant under uniform translations, rotations, and scalings of the whole formation. However, all the existing angle-constrained formation control methods are limited to undirected triangulated sensing graphs. In this paper, we propose an angle-constrained formation control approach under a Leader–First Follower sensing architecture, where the sensing graph is *directed* and *non-triangulated*. During the formation process, the control input of each agent is communication-free and based on relative positions from neighbors measured in the local reference frame. We show that the proposed distributed formation controller ensures *global exponential stability* of the desired formation. Furthermore, it is interesting to observe that the convergence rate of the whole formation is only determined by *partial* specific angles within the target formation. The effectiveness of the proposed control algorithm is illustrated by carrying out experiments both in simulation environments and on real robotic platforms.

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1. Introduction

Distributed formation shape control is the problem of how to design a distributed controller based on available local information for a group of autonomous agents to form a specific formation shape. Depending on the type of local constraints, most of the existing formation control approaches can be categorized into displacement-based (Kang & Ahn, 2015), distance-based (Anderson et al., 2008), bearing-based (Zhao & Zelazo, 2015), and mixed constraints (Kwon et al., 2020). However, both displacement constraints and bearing constraints are dependent upon the global coordinate frame, which become difficult to utilize in specific scenarios such as the indoor and underwater environments. Distance-constrained formation control has gained significant attention since it only requires measurements in local coordinate frames. Nevertheless, it is not convenient to achieve scaling control for a distance-constrained formation. Moreover, most of existing distance-based formation control approaches only achieve local convergence.

In fact, inter-edge angle constraints are independent of the global coordinate frame and render the constrained formation

the highest degree of freedom, therefore are of our interest in this paper. Angles have been employed in formation control in early references (e.g., Bishop et al. (2012), Buckley and Egerstedt (2017), Eren et al. (2003)). In Jing et al. (2019), an angle-constrained shape determination approach (angle rigidity theory) was first proposed. With the aid of angle rigidity, the authors designed a distributed control law for formation stabilization based on angle constraints, and further obtained almost global convergence in Jing and Wang (2019). In Chen et al. (2020), the authors presented a different angle rigidity theory by taking the sign of each angle into account. While requiring all angles to be defined in a common counterclockwise direction, the formation strategy in Chen et al. (2020) relaxed the measurements from relative positions to angles.

Unfortunately, all the angle-constrained approaches introduced above require the sensing graph to be undirected and triangulated, which usually cannot be satisfied in scenarios when each agent has a limited sensing capability. In addition, there is an inevitable problem of measurement mismatch with undirected sensing, so it is more practical to consider directed sensing (Mou et al., 2015). In this paper, we achieve angle-constrained formation control in the two-dimensional space under a Leader–First Follower (LFF) structured sensing graph, which is directed and non-triangulated. LFF-based formation control has been extensively studied in the literature (e.g., Mehdifar et al. (2022), Trinh et al. (2018), Yu et al. (2009)). Different from them, the formation controller proposed in this paper is based on angle constraints

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and ensures global exponential convergence. Both simulation and experimental tests are performed to support the theoretic analysis.

The main contributions of this paper are summarized as follows: (1) To the best of our knowledge, this is the first result on angle-constrained formation under directed and non-triangulated sensing graph conditions, making a significant contribution to the development of angle-constrained formation theory; (2) The developed algorithm is *nontrivial* because existing angle-constrained approaches fail to achieve formation stabilization via directed graphs; and (3) Furthermore, an interesting observation is that the convergence rate of the whole formation in our algorithm is only determined by partial specific angles within the target formation.

2. Problem formulation

We refer to a pair $(\mathcal{G}, p)$ as a framework.¹ $p = [p_1^\top, \dots, p_n^\top]^\top \in \mathbb{R}^{2n}$ is called a configuration, and $p_i \in \mathbb{R}^2$ is the coordinate in the global reference frame of vertex i . In this work, we use $(\mathcal{G}_f, p^*)$ with $\mathcal{G}_f$ to interpret the formation shape and p^* is a configuration forming the desired shape. Let $\mathcal{G}_s = (\mathcal{V}, \mathcal{E}_s)$ be the sensing graph, which characterizes the sensing ability of agents. Given a configuration p , we use the following set to specify the set of configurations that have the same shape as p :

$$\mathcal{E}(p) \triangleq \{q \in \mathbb{R}^{2n} : q = c(I_n \otimes \mathcal{R}(\theta))p + \mathbf{1}_n \otimes \xi, \\ \mathcal{R}(\theta) \in \text{SO}(2), c \in \mathbb{R} \setminus \{0\}, \xi \in \mathbb{R}^2\}, \quad (1)$$

where c is the scale factor, θ is the rotation factor, ξ is the translation factor, $\mathbf{1}_n \in \mathbb{R}^n$, I_n , and $\text{SO}(2)$ denote unit column vector, $n \times n$ identity matrix, and the special orthogonal group in $\mathbb{R}^2$, respectively.

Consider a group of n agents modeled by a single integrator dynamics:

$$\dot{p}_i(t) = u_i(t), \quad i = 1, \dots, n, \quad (2)$$

where $p_i \in \mathbb{R}^2$ and $u_i \in \mathbb{R}^2$ are the position and the control input of agent i , respectively, with respect to the global coordinate system. In this paper, we will utilize the minimally acyclic LFF structure (Yu et al., 2009) as the condition for the sensing graph $\mathcal{G}_s$, which is directed and non-triangulated. Designated agents 1 and 2 as the leader and the first follower, respectively. Without loss of generality, we make the following assumption.

Assumption 1. The directed and non-triangulated sensing graph $\mathcal{G}_s$ is constructed such that: (i) $|\mathcal{N}_1| = 0$, $|\mathcal{N}_2| = 1$, and $|\mathcal{N}_i| = 2$, $\forall i \geq 3$; (ii) If there is an edge between agents i and j , where $i < j$, the edge must be (j, i) .

Assumption 2. The target formation graph $\mathcal{G}_f = (\mathcal{V}, \mathcal{E}_f)$ contains $\mathcal{G}_s$ as a subgraph, $\mathcal{E}_f = \mathcal{E}_s \cup \{(j, k) \in \mathcal{N}_i^s, i \in \mathcal{V}\}$, and $(\mathcal{G}_f, p^*)$ is strongly nondegenerate,² where $\mathcal{N}_i^s$ denotes the neighbor of agent i in $\mathcal{G}_s$.

Remark 1. Assumption 1 indicates that $\mathcal{G}_s$ is a LFF type graph, which belongs to a class of acyclic minimally persistent graphs (Yu et al., 2009). This assumption is not restrictive as it only requires a minimum number of links for a framework to be rigid, while rigidity has been commonly used as a condition for

both sensing and formation graphs in many references (e.g., Chen et al. (2020), Jing et al. (2019)). Additionally, Assumption 2 is widely used in the existing angle-constrained formation control results. Note that we merely consider strongly non-degenerate formations in this paper. Compared with the Han et al. (2017), Lin et al. (2015) that require an additional generic assumption, Assumption 2 is milder.

Remark 2. We highlight that all the references on angle-constrained formation control require an undirected triangulated Laman graph as the sensing graph (Chen et al., 2020; Jing & Wang, 2019; Jing et al., 2019). In this study, we consider angle-constrained formation control under sensing graphs with a LFF structure, which is a milder condition and has a significantly reduced number of edges compared with undirected triangulated Laman graphs. It should be pointed out that if $\mathcal{G}_s$ contains a LFF graph as a subgraph, our formation controller will be still valid since we can always ignore redundant sensing links by comparing the agents' angle constraints and labels (assuming each agent knows the labels of its neighbors). Note that in multi-agent coordination control, each edge between two agents usually represents an information flow. Hence, our approach benefits for reducing sensing burden.

Problem 1. Given the sensing graph $\mathcal{G}_s$ and target formation $(\mathcal{G}_f, p^*)$ satisfying Assumption 1 and 2, respectively, design a distributed control protocol u_i for each agent i with dynamics (2) based on angle constraints in the target formation and relative position measurements $\{p_i^j - p_j^i, (i, j) \in \mathcal{E}_s\}$ such that p converges into $\mathcal{E}(p^*)$ asymptotically.

3. Main result

3.1. Angle constraints in target framework

Before the controller design, we will examine Assumption 2 and show how the angles in the target formation can be used to determine its shape. Given a framework $(\mathcal{G}, p)$, a signed angle α_{ijk} represents the angle rotating from the vector $(p_j - p_i)$ to the vector $(p_k - p_i)$ under the counterclockwise direction. More specifically, $\alpha_{ijk} = \arccos(b_{ij}^\top b_{ik})$ if $\text{sign}(\det(B_{ijk})) \leq 0$, and $\alpha_{ijk} = 2\pi - \arccos(b_{ij}^\top b_{ik})$ otherwise, where $B_{ijk} = [b_{ij}, b_{ik}]$ and $b_{ij} = (p_j - p_i)/\|p_j - p_i\|$, respectively. To specify the angle constraints in target formation, use $[k]$ to denote the k th triangle corresponding to the vertex added with order k . Let $\mathcal{T}_{\mathcal{G}_f} = \{(i, j, k) \in \mathcal{V}^3 : (k, i), (k, j) \in \mathcal{E}_f, i < j < k\}$, then the set of angles in the target formation to be exploited are denoted by $\mathcal{A}_{\mathcal{G}_f} = \{\dots, \alpha_{ikj}^*, \alpha_{jki}^*, \alpha_{kij}^*, \dots : (i, j, k) \in \mathcal{T}_{\mathcal{G}_f}\}$, where α_{ikj}^* , α_{jki}^* , α_{kij}^* are the signed angles of Δ_{ijk} and α_{kij}^* is further termed as follower angle.

In Chen (2022, Lemma 2), the author showed that the shape of a non-degenerate triangle Δ_{ijk} can be uniquely determined by the following *linear constraint* based on signed interior angles:

$$A_i^{[k]} p_i + A_j^{[k]} p_j + A_k^{[k]} p_k = 0, \quad (3)$$

where $A_i^{[k]} = (\sin \alpha_{kij} l_2 - \sin \alpha_{jki} \mathcal{R}^\top(\alpha_{ikj})) \in \mathbb{R}^{2 \times 2}$, $A_j^{[k]} = \sin \alpha_{jki} \mathcal{R}^\top(\alpha_{ikj}) \in \mathbb{R}^{2 \times 2}$, $A_k^{[k]} = -\sin \alpha_{kij} l_2 \in \mathbb{R}^{2 \times 2}$ are the matrices associated with Δ_{ijk} .

Lemma 1 (Li et al., 2023, Uniqueness of the target formation). Given a target formation $(\mathcal{G}_f, p^*)$ with n vertices in $\mathbb{R}^2$ satisfying Assumption 2, the shape of $(\mathcal{G}_f, p^*)$ can be uniquely determined by angle constraints (3) in $(\mathcal{G}_f, p^*)$, i.e., $q \in \mathcal{E}(p^*)$ for $q = [q_1^\top, \dots, q_n^\top]^\top \in \mathbb{R}^{2n}$ if (3) holds for all $q_i, q_j, q_k, (i, j, k) \in \mathcal{T}_{\mathcal{G}_f}$.

¹ $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is a directed graph with n vertices, where $\mathcal{E} = \{(i, j) | i, j \in \mathcal{V}, i \neq j\}$ is an edge set with directed edges and $(i, j) \in \mathcal{E}$ denotes an edge from vertex i to vertex j . We use $\mathcal{N}_i = \{j \in \mathcal{V} | (i, j) \in \mathcal{E}\}$ to denote the neighbor set of vertex i and $|\mathcal{N}_i|$ is the cardinality of $\mathcal{N}_i$.

² A framework in $\mathbb{R}^2$ is said to be strongly nondegenerate if two outgoing edges of the agent do not collinear.

Remark 3. Lemma 1 is based on signed interior angle constraints. In fact, when signed angles are utilized as sensing measurements, all the agents need to have a common understanding of the counter-clockwise direction. However, the signed angles used in this paper are constraints that can be calculated in advance by the target formation $(\mathcal{G}_f, p^*)$. Therefore, our distributed controller does not require the common counter-clockwise direction assumption.

3.2. Formation shape stabilization

Our goal is to solve Problem 1 via available sensing measurements and angle constraints in the target formation, under Assumptions 1 and 2. Note that an angle always involves three agents. According to the LFF philosophy, the leader and the first follower are not able to meet any angle constraints based on their sensing capability. Therefore, we do not apply control to the first two agents, i.e., $u_i = 0$ for $i = 1, 2$. The controllers³ for the rest of agents are designed as follows:

$$u_k = - \left(A_k^{[k]} \right)^\top \left(A_i^{[k]} e_{ki} + A_j^{[k]} e_{kj} \right), \quad k = 3, \dots, n, \quad (4)$$

where $e_{ki} = p_i - p_k$, $e_{kj} = p_j - p_k$, $A_h^{[k]}$, $h \in \{i, j, k\}$ is the matrix in the angle-induced linear constraint (3) associated with Δ_{ijk} , $\{i, j \in \mathcal{N}_k^s\}$ in $\mathcal{G}_f$, and the target formation p^* .

Remark 4. The controller (4) has the following characteristics: (1) the controller (4) is a linear⁴ and distributed strategy based on relative position measurements corresponding to the sensing graph $\mathcal{G}_s$; and (2) the controller (4) is independent of the global reference frame.

Theorem 1. Under Assumptions 1 and 2, consider an n -agent formation with dynamics (2). By implementing the distributed control law (4), the stacked vector of positions $p = [p_1^\top, \dots, p_n^\top]^\top$ converges into $p^\dagger \in \mathcal{E}(p^*)$ exponentially, where

$$p^\dagger = c^\dagger (I_n \otimes \mathcal{R}(\theta^\dagger)) p^* + \mathbf{1}_n \otimes \xi^\dagger. \quad (5)$$

Moreover, the convergence rate is solely determined by the follower angles α_{kij}^* within target framework, where $k = 3, \dots, n$, and $i, j \in \mathcal{N}_k^s$.

Proof. Note that $p_1(t)$ and $p_2(t)$ never change along the time, thus $p_1 = p_1^\dagger$, $p_2 = p_2^\dagger$. Moreover, according to (1) and (5), $p^\dagger$ has the same shape as p^* , then the angle linear constraints (3) induced from p^* are still satisfied in $p^\dagger$. Let $\tilde{p}_k = p_k - p_k^\dagger$, $k = 1, \dots, n$. In the following, we shall first establish the results for agent 3 and then extend the proof for all $3 < k \leq n$ by induction.

Step 1 ($k = 3$): From (2) and (4), we have

$$\dot{\tilde{p}}_3 = - \left(A_3^{[3]} \right)^\top \left(A_1^{[3]} e_{31} + A_2^{[3]} e_{32} \right). \quad (6)$$

Choose the Lyapunov function $V_1(t) = \frac{1}{2} \|\tilde{p}_3(t)\|^2$. The derivative of $V_1(t)$ along the trajectory of system (6) is

$$\begin{aligned} \dot{V}_1 &= -\tilde{p}_3^\top \left(A_3^{[3]} \right)^\top \left(A_1^{[3]} e_{31} + A_2^{[3]} e_{32} \right) \\ &= -\tilde{p}_3^\top \left(A_3^{[3]} \right)^\top \left(A_1^{[3]} p_1^\dagger + A_2^{[3]} p_2^\dagger + A_3^{[3]} p_3 \right) \end{aligned}$$

³ It is important to emphasize that the formation algorithm based on (Chen, 2022, Lemma 2) cannot achieve directed formation, even if it can still be implemented when the sensing graph in Chen (2022, Lemma 2) is replaced with a directed and non-triangulated graph.

⁴ In contrast to the nonlinear controller developed in Bishop et al. (2012), Buckley and Egerstedt (2017), Eren et al. (2003), Jing et al. (2019) and Jing and Wang (2019), it is convenient to apply (4) to real robotic platforms, as will be shown in Section 4.

$$\begin{aligned} &= -\tilde{p}_3^\top \left(A_3^{[3]} \right)^\top A_3^{[3]} \tilde{p}_3 \\ &= -2 \sin \alpha_{312}^2 V_1, \end{aligned} \quad (7)$$

where the third equality and fourth equality follow from the fact that $A_1^{[3]} p_1^\dagger + A_2^{[3]} p_2^\dagger + A_3^{[3]} p_3^\dagger = 0$ and $A_3^{[3]} = -\sin \alpha_{312} I_2 \in \mathbb{R}^{2 \times 2}$ in (3). Solving the differential inequality (7) on $[0, \infty)$, we have $V_1(t) = \exp(-2 \sin \alpha_{312}^2 t) V_1(0)$. Thus, $\lim_{t \rightarrow \infty} V_1(t) = 0$. That is, p_3 converges to $p_3^\dagger$ with an exponential rate $2 \sin \alpha_{312}^2$ over the time. More specifically, the equilibrium $p_3^\dagger$ is global exponential stable (GES).

Induction Step: Consider we generate the graph step-by-step in the analysis by adding a vertex k ($k \geq 4$) with two outgoing edges to any two distinct vertices i and j of the previous graph, we obtain the following cascade system at each step:

$$\dot{\tilde{p}}_k = f_k(\Pi_{k-1}, \tilde{p}_k), \quad (8a)$$

$$\dot{\Pi}_{k-1} = g_{k-1}(\Pi_{k-1}), \quad (8b)$$

where $\Pi_{k-1} := [\tilde{p}_3, \tilde{p}_4, \dots, \tilde{p}_{k-1}]$. Note that the GES of $\dot{\Pi}_{k-1} = 0$ for (8b) was already established in the previous step. Therefore, we only need to check if (8a) is input-to-state stable (ISS) with respect to input Π_{k-1} .

According to (4), the dynamics of agent $\tilde{p}_k$ is

$$\begin{aligned} \dot{\tilde{p}}_k &= - \left(A_k^{[k]} \right)^\top \left(A_i^{[k]} e_{ki} + A_j^{[k]} e_{kj} \right) \\ &= - \left(A_k^{[k]} \right)^\top \left(A_i^{[k]} \tilde{p}_i + A_j^{[k]} \tilde{p}_j + A_k^{[k]} \tilde{p}_k \right), \end{aligned} \quad (9)$$

where $\{i, j \in \mathcal{N}_k^s\}$. We consider (9) as a cascade system with $\tilde{p}_i$ and $\tilde{p}_j$ being inputs to the unforced system

$$\dot{\tilde{p}}_k = f_k(0, 0, \tilde{p}_k) = - \left(A_k^{[k]} \right)^\top A_k^{[k]} \tilde{p}_k = -\sin \alpha_{kij}^2 I_2 \tilde{p}_k. \quad (10)$$

Likewise, the unforced error system (10) is GES at $p_k^\dagger$. As a result, (8a) is ISS by Khalil (1996). Finally, we can conclude that $[\Pi_{k-1}, \tilde{p}_k] = 0$ in (8) is GES (Khalil, 1996). Repeating this process until $k = n$ leads to the conclusion that $\tilde{p} = [\tilde{p}_1^\top, \dots, \tilde{p}_n^\top]^\top = 0$ is GES, which implies $\lim_{t \rightarrow \infty} p_k = p_k^\dagger$, $k = 3, \dots, n$.

Next we derive the detailed expressions of $c^\dagger$, $\theta^\dagger$, and $\xi^\dagger$. By Lemma 1, the values of $c^\dagger$, $\theta^\dagger$, and $\xi^\dagger$ can be uniquely determined by the positions of arbitrary two agents in the formation. Note that $p_l^\dagger = c^\dagger \mathcal{R}(\theta^\dagger) p_l^* + \xi^\dagger$ ($l = 1, 2$). Let $e_{21}(0) = p_1(0) - p_2(0)$ and $e_{21}^* = p_1^* - p_2^*$. According to $p_l^\dagger = p_l(0)$, we have $e_{21}(0) = c^\dagger \mathcal{R}(\theta^\dagger) e_{21}^*$. Since the rotation matrix $\mathcal{R}(\theta^\dagger)$ multiplication has no effect on the vector size, $\|e_{21}(0)\| = c^\dagger \|e_{21}^*\|$ by the fact that $\|\mathcal{R}(\theta^\dagger)\| = 1$. Therefore, $c^\dagger = \|e_{21}(0)\| / \|e_{21}^*\|$ and $\theta^\dagger = \arccos(b_{21}^\top b_{21}^*)$ if $b_{21}^\top \mathcal{R}(\frac{\pi}{2}) b_{21}^* \geq 0$, and $\theta^\dagger = 2\pi - \arccos(b_{21}^\top b_{21}^*)$ otherwise.

Substituting $\theta^\dagger$ into $p_l^\dagger = c^\dagger \mathcal{R}(\theta^\dagger) p_l^* + \xi^\dagger$, we get $\xi^\dagger = p_1(0) - c^\dagger \mathcal{R}(\theta^\dagger) p_1^*$. ■

There is an implicit assumption in Theorem 1 that $p_1(0) \neq p_2(0)$. Since the leader and the first follower keep stationary, we assume that the first two agents do not coincide initially. There are no additional limitations for the initial positions of the rest of agents.

Remark 5. In contrast to distance-based and bearing-based methods, the angle-constrained approach is more convenient to achieve maneuver control owing to the advantage of invariance of angles in translation, rotation, and scaling motions. Similar to Chen and Sun (2022) and Jing et al. (2019), we show that applying extra reference inputs to partial agents is sufficient for maneuvering the whole formation according to Theorem 1. Due to space limitation, the detailed design of the maneuver controller is omitted here and can be found in Li et al. (2023).

Nevertheless, we will include maneuver control experiment results to validate the design methodology. Moreover, we obtain global exponential convergence in this paper, implying better performance and greater robustness to perturbations. Therefore, it is convenient to apply (4) to real robotic platforms, as will be shown in Section 4. In particular, it is interestingly noticed from (7) that the convergence rate of the formation is only determined by the follower angles $\{\sin_{kij}\}_{(i,j \in \mathcal{N}_k^s)}$ obtained by the target formation beforehand. Therefore, the convergence rate of the formation algorithm reaches maximum when each follower agent in the target formation maintains an angle of either $\pi/2$ or $3\pi/2$ with its neighboring agents.

Remark 6. As an inspiration, here we propose a single-integrator-based control law. More elaborate control laws may be required in practice to meet more complicated dynamics. In fact, our reasoning remains valid for such control laws provided the notion of equilibrium is appropriately redefined based on connections between different dynamic models. For example, in order to control the differential-drive robots to achieve specific formation, single-integrator dynamics can always be mapped to unicycle models through a near-identity diffeomorphism (NID) (Wilson et al., 2020).

Avoiding collision among agents is an important issue in practical formation control (Fang et al., 2023). In this regard, an algebraic condition for collision avoidance among triangles in $\mathcal{G}_f$ can be derived if the control law (4) is implemented for each agent i after its two neighboring agents are stabilized, see the following lemma.

Lemma 2 (Li et al., 2023 Collision-Free Between Agent i and Its Neighbors). Suppose the control law (4) for each agent $i \geq 3$ is implemented after its two neighboring agents are stabilized. Agent i never collides with its neighboring agents if $\|p_i(0) - p_i^*\| < \min\{\|p_i^* - p_j^*\|, \|p_i^* - p_k^*\|\}$.

4. Experiments

4.1. Formation shape stabilization

In this subsection, we validate the effectiveness of Theorem 1 using a team of differential-drive robots via the “Robotarium” platform⁵ (Wilson et al., 2020). Consider the directed non-triangulated sensing graph and the target formation $(\mathcal{G}_f, p^*)$ are shown in Fig. 1(a) and Fig. 1(b) respectively satisfying Assumptions 1 and 2. The set of angles in the target formation to be exploited is given as $\mathcal{A}_{\mathcal{G}_f} = \{\alpha_{123}^* = 90^\circ, \alpha_{213}^* = 315^\circ, \alpha_{324}^* = 45^\circ, \alpha_{423}^* = 270^\circ, \alpha_{415}^* = 63.43^\circ, \alpha_{514}^* = 306.87^\circ, \alpha_{416}^* = 296.57^\circ, \alpha_{614}^* = 53.13^\circ\}$. To fit the 12×14 -ft² testbed, the initial positions of six differential-drive robots are chosen as $p(0) = [0, 0.5, -0.5, 0, 0, 0.025, 0.25, 0.4, 0.5, -0.35, 0.6, 0.2]^\top$ for better visual presentation, which are shown in Fig. 1(c). The posture vectors are randomly chosen in the two-dimensional space. By implementing the formation shape control law (4), the eventual positions of robots and the corresponding evolution of angle errors in $\mathcal{A}_{\mathcal{G}_f}$ are presented respectively in Figs. 1(d) and 1(e), showing that the desired formation shape is asymptotically achieved.

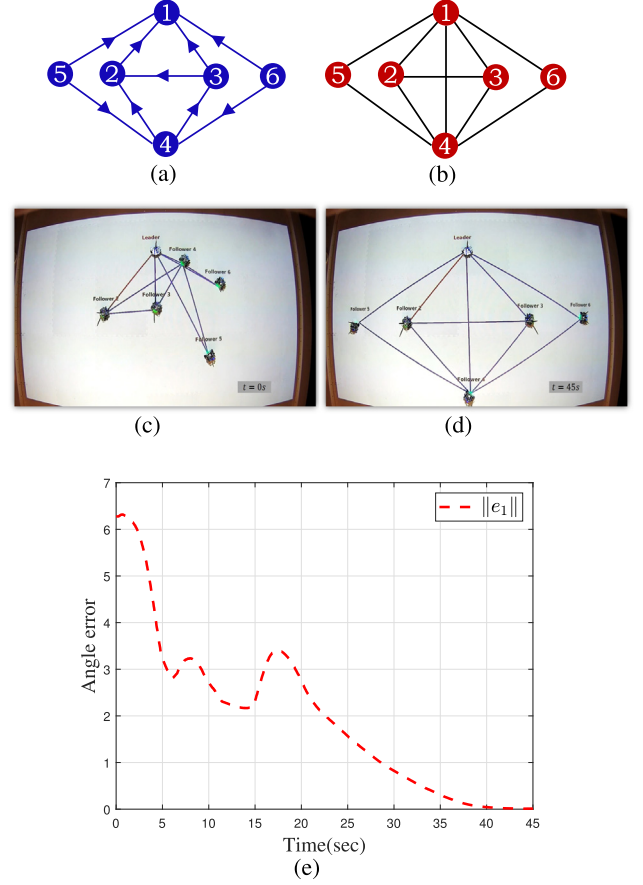


Fig. 1. The formation shape control law (4) was implemented on a team of six differential drive robots. (a) the sensing graph $\mathcal{G}_s$; (b) the target formation $(\mathcal{G}_f, p^*)$; (c) initial positions of six robots on the Robotarium; (d) the six robots eventually form the target formation shape; (e) the angle errors $\|e_1\| = \sum_{\alpha_{ijk} \in \mathcal{A}_{\mathcal{G}_f}} \|\alpha_{ijk} - \alpha_{ijk}^*\|$ converge to zero asymptotically.

4.2. Formation maneuver control

This example aims to steer the six differential-drive robots to perform a maneuver task cooperatively. The set of desired angle constraints is the same as the one used in the last subsection. The initial positions are set as $p(0) = [-0.4, -0.35, -1.1, -0.4, 0, 0.025, 0.25, 0.4, 0.5, -0.35, 0.6, 0.2]^\top$, which are shown in Fig. 2(a). The posture vectors are randomly chosen from the two-dimensional space. The desired piecewise-constant translational velocities are set as $v_r^*(t) = [0, 0.02]^\top, t \in [0, 50]$; $v_r^*(t) = [0.05, 0]^\top, t \in (50, 90]$; $v_r^*(t) = [0.04, 0]^\top, t \in (90, 120]$, where $v_r^*(t)$ denotes the target velocity of the formation; The piecewise-constant relative positions between the leader and the first follower are prescribed as $\delta_{12}^*(t) = [0.4, 0.4]^\top, t \in [0, 50]$; $\delta_{12}^*(t) = \mathcal{R}(-\frac{\pi}{2})[0.4, 0.4]^\top, t \in (50, 90]$; $\delta_{12}^*(t) = 0.7 * \mathcal{R}(-\frac{\pi}{2})[0.4, 0.4]^\top, t \in (90, 120]$. More detailed information about these variables can be found in Li et al. (2023). By implementing the maneuver control law, the snapshots of the formation in different time intervals and the corresponding evolution of angle errors in $\mathcal{A}_{\mathcal{G}_f}$ are presented respectively in Fig. 2, showing that the formation maneuvering with translation, rotation and scaling is achieved.

⁵ An illustrative simulation video is available at: <https://youtu.be/LUCyMYFR7y8>.

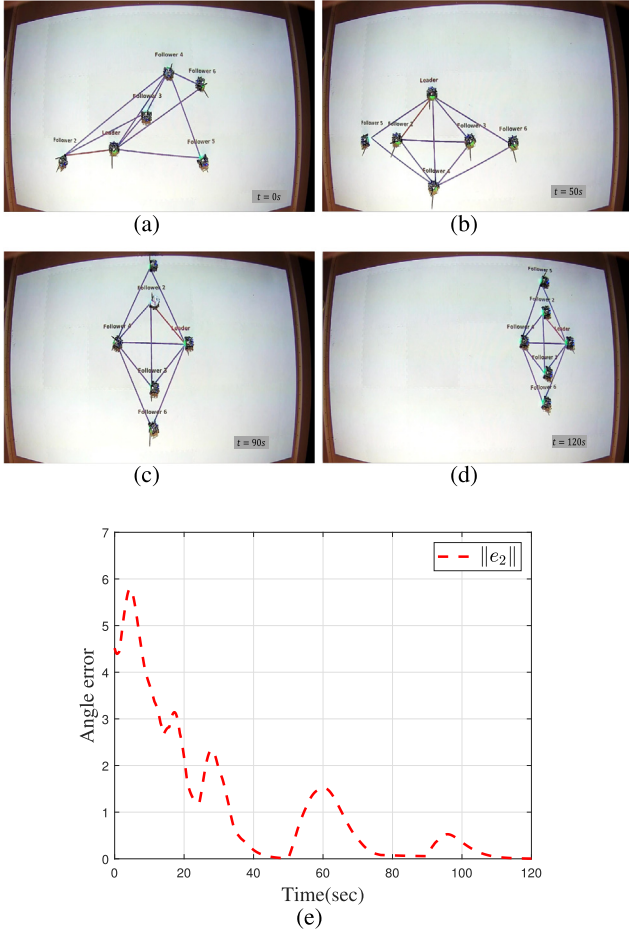


Fig. 2. The formation maneuver control law was implemented on a team of six differential drive robots. (a) initial positions of six differential-drive robots on the Robotarium; (b) the formation at time $t = 50$ s; (c) the formation at time $t = 90$ s; (d) the formation at time $t = 120$ s; (e) evolution of angle errors $\|e_2\| = \sum_{\alpha_{ijk} \in \mathcal{A}_{G_f}} \|\alpha_{ijk} - \alpha_{ijk}^n\|$.

5. Conclusion

In this paper, we achieved angle-constrained formation shape stabilization and maneuver control under a directed and non-triangulated sensing graph. The developed distributed controller can be implemented in the local reference frame and ensure global convergence of angle errors. Future work may include angle-constrained formation control with directed sensing graphs in three-dimensional space and angle-constrained formation control with more general graphs.

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